

Under five minutes, **ten slides.**

What to say on every slide, **29 September.**

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Run of show · 29 Sept

Under five minutes, ten slides: 460 spoken words and 15 clicks, 3:38 at 150 words a minute. Detail that is not on a slide is in the notes here. The spine: **self, us, now** (02, 03), then **why, how, what** (04, 05, 06), then **who else, where it goes, the ask** (07–10). 07 and 08 are the fastest stretch: one beat each.

SLIDE	CLOCK	SEC	CLICKS	BEAT	ONE LINE TO LAND
01 Control is an illusion	0:00–0:11	12	1	hook	Confidence without control.
02 Founder	0:11–0:39	28	2	self	This is the thing I kept needing and nobody sold.
03 Do my cells look normal?	0:39–1:02	22	1	us, now	Cells are the input nobody certifies.
04 One lab, or six?	1:02–1:33	32	3	why now	The model isn't the hard part. The reference is.
05 It passed every check	1:33–2:14	40	4	how	Your own past is not a reference. The field is.
06 The product	2:14–2:36	22	1	what	Four months becomes one day.
07 Identity, yes. Behaviour, no.	2:36–2:46	10	1	who else	The other half of the certificate.

SLIDE	CLOCK	SEC	CLICKS	BEAT	ONE LINE TO LAND
08 Three buyers, one certificate	2:46–3:01	15	1	who buys	Contributors use it free.
09 The ugliest dataset in biology	3:01–3:16	15	1	the build	Name the failure, not just flag it.
10 Every plate, placed in the field	3:16–3:38	22	0	vision, ask	Have images of U2OS cells? Talk to me.

Captions stay hidden while you present (Tweaks › captions); the shared link and PDF show them. If you're late, skip the fourth view of 05 and the first sentence of 10.

01 Control is an illusion (0:00–0:11)

On screen while you're introduced. Two lines drop from the top like dive lines: green is the field's datum, pink is yours, and the gap between them is the product. Let the dial drift.

0:00 Hi, I'm Kat. Every result in biology is a difference from a control, and nobody checks the control. So I'm building confidence without it. *Click.*

02 Founder (0:11–0:39)

Self. Let the logos and the pink line under your name do the credential work.

0:11 Five labs, and the same question at every bench: do my cells look normal? Nobody could answer it. *Click.*

0:20 At Axiom we onboarded hepatocytes through a good CRO. Their checks passed; ours passed. The first sign anything had moved came from outside: the Broad couldn't align our profiles to their reference.

0:33 This is the thing I kept needing and nobody sold. *Click.*

Founder detail, for the judge question (about 30 seconds). Three methods, three papers, one thread. At AbbVie I built a neuronal tau aggregation assay for drug discovery [3]: the readout was what the cells did, not a label on them. At Axiom I built Cell Painting in primary human hepatocytes for cytotoxicity and mode of action [1], ran assay development for two of the largest Cell Painting datasets in the world, and invented two bioactivity detection methods that see a small molecule's effect on cells at about ten times lower concentration than LDH, ATP or cell count. With the Carpenter lab we showed that cell count alone predicts small-molecule bioactivity benchmarks [2]: even the simplest phenotype carries the signal. The thread is that bioactivity is a measurement of behaviour, and behaviour is the half of QC nobody measures. Every lab already images its control wells; Datum Line turns that image into the certificate, against the field.

Precision to hold. The question predates plates: at MGH it was whether a *Xenopus* oocyte, a single frog egg cell, looked normal before a recording. Say only what your author contribution statements support: "built" on [1] and [3], "we showed" on [2]. The two bioactivity methods and the ten-fold sensitivity are from your resume (macrofluidic.me/resume); if asked, name the comparison assays and the concentration range. The 2019 GABA_A paper [5] is the MGH anesthesia lab, Harvard Medical School, 2017 to 2018. The industry claim is the deck's own: STR and vendor certificates cover identity, bioactivity covers behaviour, and behaviour is what a control well measures in any lab that images one.

Precision to hold. "Published with" lists coauthor institutions, never partners. The Axiom story has no villain: the CRO passed every check it had, Axiom passed every check it had, and the failure was that no check for this existed. Name no vendor and no person. The uneven distribution was noticed; its severity and its effect on data quality were unknown, so never present it as the cause, and never say "spun down unevenly". If asked what it was: "We still don't know for certain. The uneven spreading is one hypothesis; whole-well images show it more easily than our fields, and a field reference is what would tell you whether it matters." The sequence is: local checks passed, uneven spreading noticed but unquantified, the Broad flagged that reference

profiles would not align, the cause is still open. Say what happened, not how it was fixed. Confirm with Axiom that you can tell it; if not, say "at one lab".

03 Do my cells look normal? (0:39–1:02)

Us, then now. Point at the pink slice, then the six search results, then the two edges of card 3: run it, or hold.

0:39 It's not just me. About twenty-eight billion dollars a year of US preclinical research can't be reproduced, and a third of that is inputs.

0:49 Today you search, you ask around, then you guess. I've seen one lab run too fast and one hold too long. Cells are the input nobody certifies. *Click.*

Precision to hold. \$28B is a 2015 US estimate; the 36% covers all reagents and reference materials, not only cells. The search card shows six real U2OS images from six papers, 2009 to 2023, via IDR and JUMP; sources are in the README. Card 3 names neither lab and gives no exact figure; keep it that way on stage. Growth chart is the metaphor if you need one: nobody plots their cells against the population.

04 One lab, or six? (1:02–1:33, three views)

Why now. The first view is a game: let the room look for four seconds.

1:02 Six untreated wells. Same line, same protocol, no drug. One lab, or six? *Click.*

1:09 Six. Nobody here can tell. A classifier can: 99.9 percent, against nine by chance, on plates it never saw. *Click.*

1:18 Why now: this data is public and free, and models are a commodity. The model isn't the hard part. The reference is. *Click.*

Precision to hold. Eleven labs, 93,228 untreated wells, 1,872 plates, held out by plate; 78 wells wrong. Holding out whole batches gives 99.6%; labs on one

microscope model, 99.99%. The signature is in every feature family, image-quality features included, so it reflects the whole imaging pipeline as well as the cells. Say "the lab", not "the cells". Cell Painting fluorescence; brightfield is the bet.

For ML people. The line on the second view: normalising each plate to its own controls takes lab identity from 99.9 to 24 percent. Right for discovery, wrong for QC (A3).

05 It passed every check (1:33–2:14, four views)

How. Dark slide: slow down. Four views of one lab, one sentence each.

1:33 One lab, thirteen batches, June to October. Cell counts stay in range. *Click.*

1:40 Now the same batches against the field. At the end of August the lines cross: this lab is further from its own June baseline than from another lab. *Click.*

1:53 Same lab, two different places. Your own past is not a reference. The field is. *Click.*

2:01 And it matters. The further a batch drifts, the less its answer to the same drugs agrees with its own baseline. That's the failure I lived. *Click.*

Figure detail. Second view: median distance of each batch's plates from the lab's first batch, and to the nearest other lab's centre, in units of the field's within-plate spread; features scaled on the other 10 labs, 82,319 wells, 3,773 features, clipped at five standard deviations. By October: 8.1 from itself, 5.1 from lab 8; every well is still named lab 2. On 21 June the cell count dips to 150, which the lab can see. Third view: a PCA of 15,352 wells for the eye; distances are measured in all features. Two plates of this lab sit 0.63 spreads apart in one batch and 4.7 after thirteen weeks, seven times further. Fourth view: one point per batch; agreement is the cosine between a batch's effect of 7 positive controls and its lab's first-batch effect; rho is Spearman after removing each lab's mean, minus 0.71 over 118 batches; shuffled, minus 0.03;

without lab 2, minus 0.68; drift and effect from different wells, minus 0.72. Lab 2 against its June answer: 0.85 before the flag, 0.67 after.

Precision to hold. In lab 2 the flagged batches are also the last three, so inside that lab the flag and the calendar cannot be separated; the correlation uses every lab. The crosses on the fourth view are batches whose positive controls show no effect (A8); they are not drift. If you are late, skip the fourth view.

06 Know on day one, not month four (2:14–2:36)

What. One sentence, then the strip left to right, then the certificate.

2:14 The product: send the control images you already take. They stay yours. One answer in a day.

2:20 Here it is, replayed on that lab. Nine batches in. On 30 August, out: five weeks before October. And it says where to look: stain intensity, not cell count. Four months becomes one day. *Click.*

Precision to hold. The replay is hindsight on public data. The flagged batches hold 5, 6 and 14 plates. The certificate says where to look, not why. Across the field 6 of 129 batches flag, and lab 2's verdicts hold under nine of ten variants (A11, A12). Brightfield is still the bet (A10).

Figure detail. Each batch is judged with only its own lab's history up to that day, against the field as published. Outside means further from its own first batch than from the nearest other lab. The certificate is the first flag, 30 August: 5 plates, 13.0 spreads from baseline, 8.9 to the nearest lab, 165 cells per well. ER and AGP stain intensity are up 2.9 and 2.8 standard deviations, Mito 1.4.

07 Identity, yes. Behaviour, no. (2:36–2:46)

Point at the open tick in the Datum Line column: identity is done with a partner, STR. One sentence.

2:36 STR proves which line you have. Nothing proves it behaves like everyone else's. STR is a collaborator, not a rival. *Click.*

08 Three buyers, one certificate (2:46–3:01)

Who buys. Walk the three columns; the pink row is what they pay.

2:46 Three buyers. Cores contribute controls and use it free: that builds the reference. Pharma and CROs pay per line, per quarter: a fraction of one unchecked screen. Cell banks pay per certified lot. *Click.*

Precision to hold. The prices are a proposal, not a validated model: free for contributors, about 5 to 15 thousand dollars per line per quarter for screening groups, a fee per certified lot. One unchecked screen is about 2.6 million, so the subscription is about 170 times less. Bad-lot alerts come after the certificate.

09 The ugliest dataset in biology (3:01–3:16)

The build. This is the blank-cheque answer, said before anyone asks. Point at the six real wells, then the four corrupted tiles.

3:01 With a blank cheque I'd build the ugliest dataset in biology, on purpose. Six labs already look this different. Then plant the failures, and make every method name which one went wrong. *Click.*

Figure detail. Six real untreated wells, one per lab, with each lab's instrument; across the 13 JUMP labs there are five microscope models, objective NA from 0.4 to 1.0, and eight plate types. The four red-team tiles are one real well corrupted in software and are labelled simulated. The failures to plant: bubble, focus, exposure, clipped well, bad lot, wrong label.

10 Every plate, placed in the field (3:16–3:38)

Vision, then the ask. Point at the grid, then the right card. Stop on the question.

3:16 The vision: every plate you image, placed in the field. Here it is on public data: 129 batches, eleven labs. Six outside their baseline. Four where a known answer failed.

3:28 What I need: images, U2OS first; ML people better than me; bench time; and a pre-seed.

3:34 Have images of U2OS cells? Talk to me.

Precision to hold. Price and test-set budget are not set: say "per line, per quarter" and "the proof of concept sets the size". Don't name a partner lab until one has agreed. Name no JUMP lab beyond its number. The four failed batches are one lab's, 65 plates; the data cannot say whether the compounds were left out or the plate map is wrong, so say neither. Next it tells you what to change; then it flags a bad lot across every lab using it.